Course Information Sheet

Discrete Mathematics (ICT1123)

Document No. CRS-0040 · Academic Year 2024/2025 · Semester 2

Course Details

Field	Details
Course ID	2
Course Name	Discrete Mathematics
Course Code	ICT1123
Credit Value	3.00
Department	Information and Communication Technology (Department ID: 1)

Course Description

This course presents the mathematical foundations required for computing and information technology. Topics include propositional and predicate logic, proof techniques, sets, relations, functions, combinatorics, and an introduction to graph theory. Throughout the course, abstract concepts are linked to computing applications such as database theory, algorithm analysis, and digital logic. Students develop the rigor needed to reason precisely about the correctness and complexity of computational processes.

Learning Outcomes

On successful completion of this course, students will be able to:

- Formulate statements in propositional and predicate logic and construct valid proofs.
- Manipulate sets, relations, and functions and explain their role in computing.
- Apply counting techniques and elementary combinatorics to practical problems.
- Model simple problems using graphs and trees and reason about their properties.

Assessment and Grading

Component	Weighting	Description
Assignments and Quizzes	20%	Graded coursework issued and completed during the semester
Mid-Semester Test	10%	Written test covering the first half of the course content
Final Examination	70%	End-of-semester written examination

Note: this is a sample document generated for project testing purposes. Assessment weights and descriptive text are illustrative.